INNOFUSION 2.0

KOLKATA'S PREMIER SOFTWARE + HARDWARE HACKATHON

8TH – 9TH MARCH, 2025

Team Name: Neural Nexus

Name of College/University: Ramdeobaba University

Team Member Details:

- Pratik Tiwari
- Vidhi Umap
- Vedant Sikariya

Problem Statement:

In large-scale institutions such as schools or daycare centers, tracking children's presence and movement accurately is a major challenge. Manual attendance systems are time-consuming, error-prone, and unreliable in ensuring safety. There’s also no real-time awareness about a child’s location within or outside the designated campus zone. This can lead to safety concerns or potential emergencies going unnoticed.

Understanding of the Problem:

We understood that the core issue revolves around the lack of an automated system to:

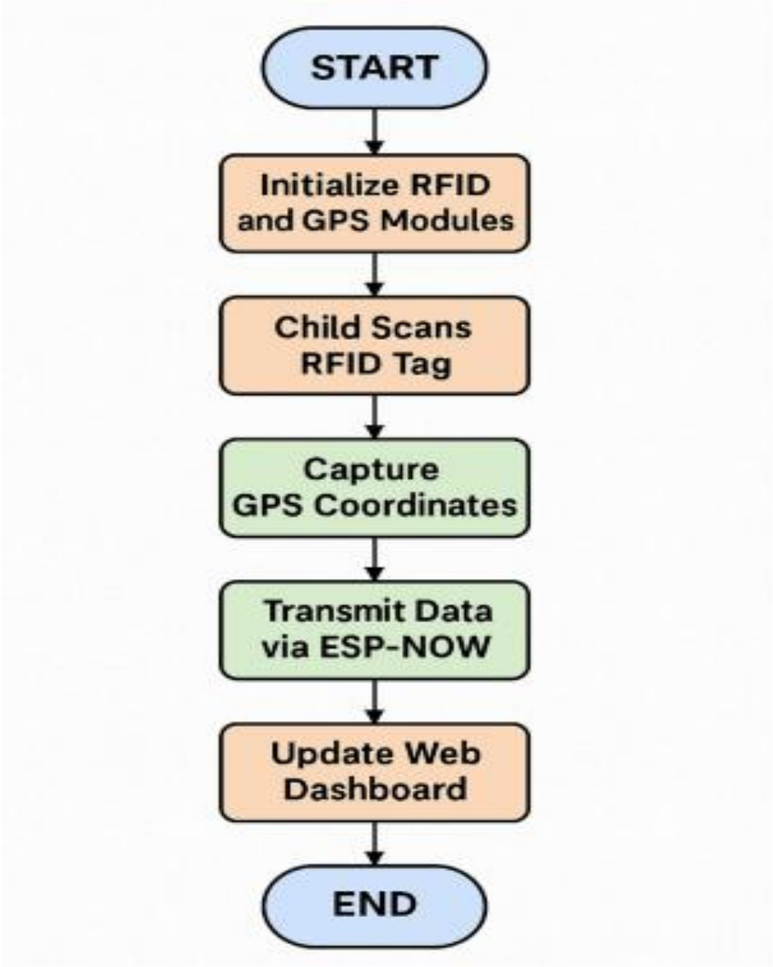
1. **Identify each child individually**
2. **Log attendance securely**
3. **Track live location with boundary alerts**
4. **Visualize the data on a live platform accessible to authorities**

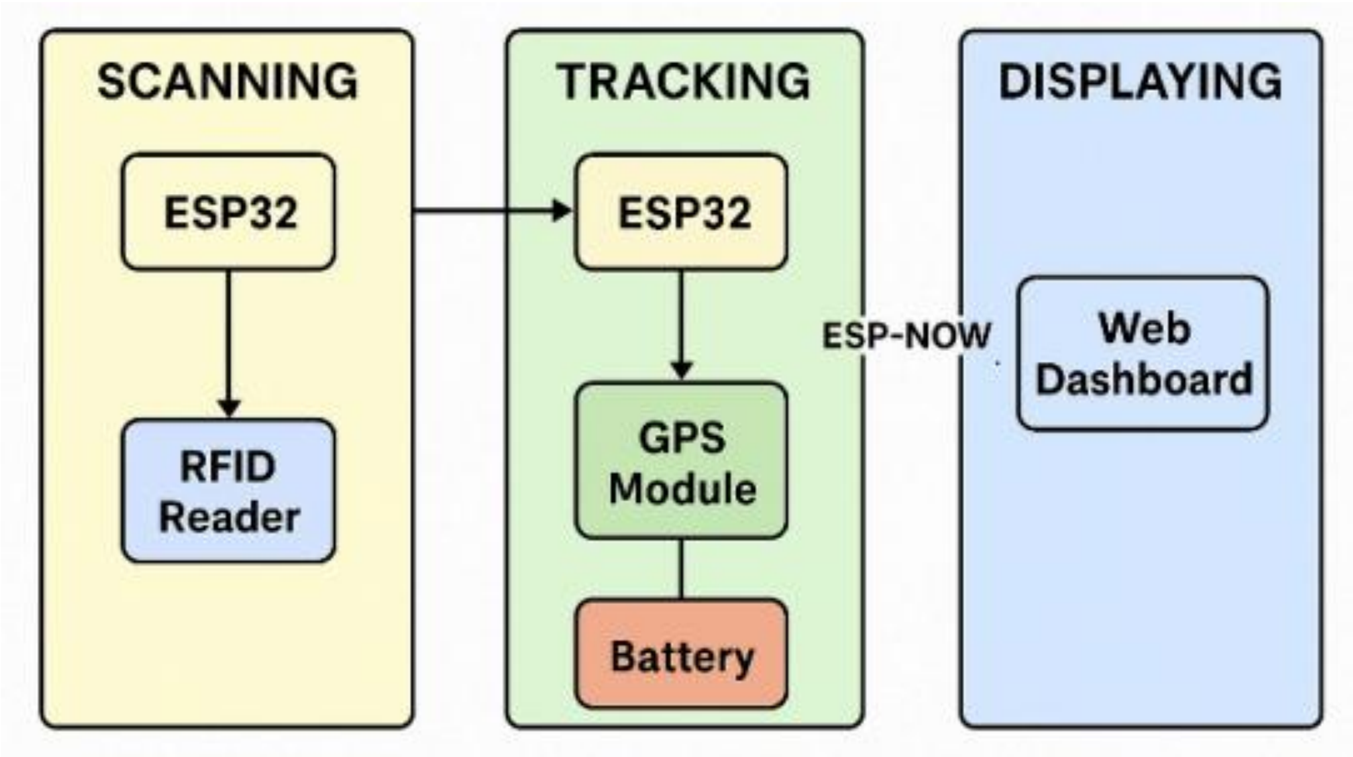
A scalable, tech-driven system can address all of this efficiently with minimal manual intervention.

Brief About Our Approach:

We propose a hybrid IoT-based system combining **RFID**, **GPS**, **Firebase**, and **wireless protocols**. Each child carries an RFID-based identity card and a GPS tracking unit that communicates live location data. Scanning stations mark attendance and verify identity, while the GPS unit transmits coordinates to a central ESP32, which pushes data to Firebase and displays the location on Google Maps.

Flowchart & Working:





Detailed Proposal & Solution Approach

System Overview:

- **RFID Scanning System**

Child scans RFID tag on an ESP32-based scanning bench.
Marks attendance & stores child's details
(Name, Contact, Parents' Info, Address).

- **Live GPS Tracking**

Child carries an ESP32-based GPS tracker
(GPS module + Battery Module).
Live coordinates sent via ESP-NOW to another ESP32.

- **Data Storage & Dashboard**

GPS data stored in Firebase.
Real-time location displayed on a web dashboard/Mobile App.

Tools and devices used on development:

Tool / Device	Use
ESP32 Dev Board	Microcontroller for both scanning and tracking
RC522 RFID Reader	Tag-based attendance
RFID Cards	Unique ID for each child
NEO-6M GPS Module	Location tracking
Li-ion Battery + TP4056	Power supply for portable tracker
LEDs + Buzzer	Alerts on scanning or boundary breach

Technologies involved:

Technology	Purpose
ESP-NOW Protocol	Wireless data transmission (short range)
Firebase Realtime DB	Cloud storage for tracking data
Google Maps API	Display real-time location on map
OpenCV (future scope)	Face recognition-based identity (optional)
HTML/CSS + JS	Web dashboard UI

References:

[1] S. K. Singh et al., “Real-time school bus tracking system using GPS and IoT.” IEEE Trans. Edu. Tech., 2020.
[2] A. Gupta, “RFID based attendance system,” IJETT, vol. 68, no. 3, 2020.
[3] P. Das et al., “Smart child safety wearable,” IEEE Conf. on IoT, 2019.
[4] M. Smith, “ESP32 for IoT solutions,” Embedded Journal, 2021.
[5] J. Patel, “Wireless child monitoring using ESP-NOW,” IJARCET, 2022.
[6] L. Zhang et al., “Low-power GPS for smart ID systems,” IEEE Sensors J., 2020.
[7] A. S. K. Reddy, “Child safety and attendance management system using RFID and GPS,” in Proc. 2021 IEEE International
[8] K. Elissa, “GPS data transmission in urban IoT,” IEEE IoT Mag., 2021.
[9] A. Kumar, “Firebase for real-time apps,” IJCA, 2020.
[10] P. S. R. Reddy and M. S. R. Reddy, “An efficient RFID-based attendance system for schools,” in Proc. 2022 IEEE International
[11] S. Pradhan, “Data security in ESP-NOW,” IEEE IoT Trans., 2022.
[12] D. Rao, “Smart school systems,” IJSETR, 2021.
[13] H. Ali, “RFID tech in education,” IEEE Trans. Edu., 2019.
[14] N. Verma, “Child tracker based on ESP32,” IJIRST, 2022. [15] P. Sharma, “IoT safety solutions,” IJEECS, 2020.
[15] M. A. Al-Mansoori, “Design and implementation of an RFID-based attendance system,” in Proc. 2018 IEEE International Conference on Computer and Communication Engineering (ICCCE), Kuala Lumpur, Malaysia, 2018,

Links:

Firebase Documentation:
https://firebase.google.com/docs

Arduino IDE Download:
•https://www.arduino.cc/en/software

ESP32 Official Page (Espressif):
https://www.espressif.com/en/products/socs/esp32

ESP-NOW Documentation (Espressif IDF):
https://docs.espressif.com/projects/esp-idf/en/latest/esp32/api-reference/wifi/esp_now.html

U-blox NEO-6 Series:
https://www.u-blox.com/en/product/neo-6-series